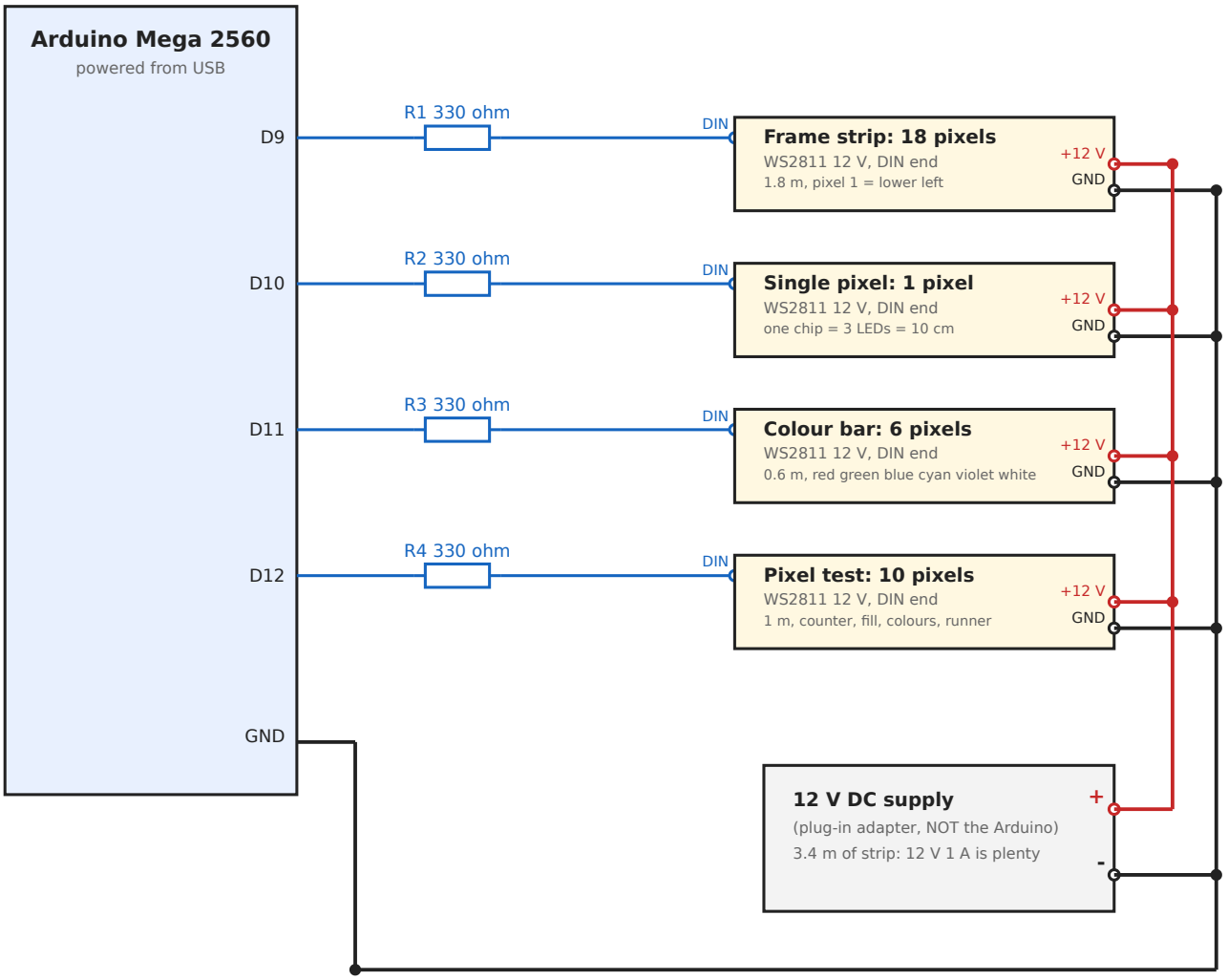
Strip frame demo test - wiring (Arduino Mega 2560 + four 12 V WS2811 strip pieces)



Connections

1. Arduino D9 -> 330 ohm resistor R1 -> DIN of the frame strip (18 pixels, the end where the printed arrows point away from).
2. Arduino D10 -> 330 ohm resistor R2 -> DIN of the single-pixel piece (one chip, 3 LEDs, cut from the same roll).

2b. Arduino D11 -> 330 ohm resistor R3 -> DIN of the colour bar (6 pixels = 18 LEDs, 60 cm, cut from the same roll).

2c. Arduino D12 -> 330 ohm resistor R4 -> DIN of the pixel test strip (10 pixels = 30 LEDs, 1 m, cut from the same roll).

3. The +12 V of all four strips comes from the 12 V supply, NEVER from an Arduino pin. No 12 V on any Arduino pin.
4. GND of the 12 V supply, GND of all strips and GND of the Arduino are connected together (common ground).
5. The Arduino is powered from USB (the Serial Monitor, 115200 baud, shows one DBG line per second).
6. Optional: a 1000 uF capacitor rated 25 V or more across +12 V and GND at the strip start (plus to +12 V).
7. The built-in LED (pin 13) blinks while the program runs.
8. Brightness peaks at 80 of 255 in the sketch (TEST_LEVEL); at most about 30 lit pixels need very little current.